

HARDEN & TEMPERED STEEL

CARBON STEEL KS D3752, JIS G4051

KS JIS	AISI	Chemical Composition %					Mechanical Properties					
		C	Si	Mn	P	S	YP	TS	EL	RA	Impact	Hardness
							kg/mm ²	kg/mm ²	%	%	charpy	HB
S10C	1010	.08/.13 .08/.13	.15/.35 -	.30/.60 .30/.60	≤0.03 ≤.035	≤.035 ≤.045	≥21	≥32	≥33			109/159
S12C	1012	.10/.15 .10/.15	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥24	≥38	≥30	-	-	111/167
S15C	1015	.13/.18 .13/.18	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥24	≥38	≥30	-	-	111/167
S17C	1017	.15/.20 .15/.20	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥25	≥41	≥28	-	-	116/174
S20C	1020	.18/.23 .18/.23	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥25	≥41	≥28	-	-	116/174
S22C	1023	.20/.25 .20/.25	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥27	≥45	≥27	-	-	123/183
S25C	1025	.22/.28 .22/.28	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥27	≥45	≥27	-	-	123/183
S28C	1029	.25/.31 .25/.31	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥29	≥48	≥25	-	-	137/197
S30C	1030	.27/.33 .28/.34	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥29	≥48	≥25	-	-	137/197
S33C		.30/.36	.15/.35	.60/.90	≤.030	≤.035	≥31	≥52	≥23	-	-	149/207
S35C	1035	.32/.36 .32/.38	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥31	≥52	≥23	-	-	149/207
S38C	1038	.35/.41 .35/.42	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.030	≤.035 ≤.045	≥33	≥55	≥22	-	-	156/217
S40C	1040	.37/.43 .37/.44	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.030	≤.035 ≤.045	≥33	≥55	≥22	-	-	156/217
S43C	1043	.40/.46 .40/.47	.15/.35 -	.60/.90 .70/1.00	≤.030 ≤.035	≤.035 ≤.045	≥35	≥58	≥20	-	-	167/229
S45C	1045	.42/.48 .43/.50	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥35	≥58	≥20	-	-	167/229
S48C		.45/.51	.15/.35	.60/.90	≤.030	≤.035	≥37	≥62	≥18	-	-	179/235
S50C	1049	.47/.53 .46/.53	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥37	≥62	≥18	-	-	179/235
S53C	1053	.50/.56 .48/.55	.15/.35 -	.60/.90 .70/1.00	≤.030 ≤.035	≤.035 ≤.045	≥40	≥66	≥15	-	-	183/255
S55C	1055	.52/.58 .50/.60	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥40	≥66	≥15	-	-	183/255
S58C	1060	.55/.61 .55/.65	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥40	≥66	≥15	-	-	183/255